

Systems biology

TGSA: protein–protein association-based twin graph neural networks for drug response prediction with similarity augmentation

Yiheng Zhu ^{1,†}, Zhenqiu Ouyang ^{2,†}, Wenbo Chen^{2,1}, Ruiwei Feng¹, Danny Z. Chen ³, Ji Cao^{4,*} and Jian Wu^{5,*}

¹College of Computer Science and Technology, Zhejiang University, Hangzhou 310000, China, ²Polytechnic Institute, Zhejiang University, Hangzhou 310000, China, ³Department of Computer Science and Engineering, University of Notre Dame, Notre Dame, IN 46556, USA, ⁴College of Pharmaceutical Sciences, Zhejiang University, Hangzhou 310000, China and ⁵Department of Ophthalmology of the Second Affiliated Hospital School of Medicine, and School of Public Health, Zhejiang University, Hangzhou 310000, China

*To whom correspondence should be addressed.

[†]The authors wish it to be known that, in their opinion, the first two authors should be regarded as Joint First Authors.

Associate Editor: Pier Luigi Martelli

Received on June 17, 2021; revised on August 16, 2021; editorial decision on September 1, 2021; accepted on September 24, 2021

Abstract

Motivation: Drug response prediction (DRP) plays an important role in precision medicine (e.g. for cancer analysis and treatment). Recent advances in deep learning algorithms make it possible to predict drug responses accurately based on genetic profiles. However, existing methods ignore the potential relationships among genes. In addition, similarity among cell lines/drugs was rarely considered explicitly.

Results: We propose a novel DRP framework, called TGSA, to make better use of prior domain knowledge. TGSA consists of Twin Graph neural networks for Drug Response Prediction (TGDRP) and a Similarity Augmentation (SA) module to fuse fine-grained and coarse-grained information. Specifically, TGDRP abstracts cell lines as graphs based on STRING protein–protein association networks and uses Graph Neural Networks (GNNs) for representation learning. SA views DRP as an edge regression problem on a heterogeneous graph and utilizes GNNs to smooth the representations of similar cell lines/drugs. Besides, we introduce an auxiliary pre-training strategy to remedy the identified limitations of scarce data and poor out-of-distribution generalization. Extensive experiments on the GDSC2 dataset demonstrate that our TGSA consistently outperforms all the state-of-the-art baselines under various experimental settings. We further evaluate the effectiveness and contributions of each component of TGSA via ablation experiments. The promising performance of TGSA shows enormous potential for clinical applications in precision medicine.

Availability and implementation: The source code is available at <https://github.com/violet-sto/TGSA>.

Contact: wujian2000@zju.edu.cn or caoji88@zju.edu.cn

Supplementary information: [Supplementary data](#) are available at *Bioinformatics* online.

1 Introduction

Precision medicine aims to tailor therapeutic regimens for each patient based on individual genes, lifestyle and environment (Hodson, 2016). Although genomics tends to be synonymous with precision medicine in cancer analysis, it is still challenging to fulfill precision medicine based on genetic information, since the exact relationships between genotype and phenotype are not yet clear (Friedman *et al.*, 2015). Such challenges have prompted researchers to undertake large-scale anti-cancer drug screens and propose computational

methods to explore the potential impacts of patients' genetic profiles on drug response (Baptista *et al.*, 2021).

Recently, the development of high-throughput screening technologies has facilitated screening hundreds of anti-cancer drugs against a panel of cancer cell lines. Several large-scale cancer genetic projects have been launched and systematized into public repositories, including Cancer Cell Line Encyclopedia (CCLE) (Barretina *et al.*, 2012) and Genetics of Drug Sensitivity in Cancer (GDSC) (Yang *et al.*, 2013). Given that these datasets provide not only abundant anti-cancer drug response information but also genetic profiles of

cancer cell lines, they are of great help in spurring the development of drug response prediction (DRP) techniques.

In the past decade, two main categories of computational DRP methods have been proposed. (i) Matrix factorization: These methods generally decompose a drug response matrix into two low-rank matrices indicating the representations of cell lines and drugs. By extending matrix factorization, pathway-drug response associations (Ammad-ud din *et al.*, 2016) and cell line/drug similarity (Suphavitai *et al.*, 2018; Wang *et al.*, 2017) are learned to improve the performance. While these early attempts have made remarkable progress in DRP, they did not explore the features of drugs and cell lines and showed poor extrapolation when inferring drug response on unseen drugs or cell lines. (ii) Machine learning: many studies attempted to resolve the aforementioned issues with machine learning algorithms, e.g. elastic net (Barretina *et al.*, 2012), support vector machine (Dong *et al.*, 2015) and random forest (Iorio *et al.*, 2016). While these methods showed considerable effect, they still have major drawbacks. First, commonly used features such as molecular fingerprints of drugs and genetic profiles of cell lines are high-dimensional data, which may cause the methods to become severely overfitting (Teschendorff, 2019). Second, feature selection is indispensable but requires deep insight into the involved biological processes (Mayr *et al.*, 2016). Third, the relationships in biological data are too intricate to be modeled completely by these methods with limited capacity.

With an unprecedented power of representing high-dimensional data (e.g. images, videos and text), deep learning methods have attracted a multitude of attention. Recently, deep learning also found its way into DRP and outperformed traditional machine learning algorithms (Baptista *et al.*, 2021). According to the training modes, deep learning-based DRP methods can be categorized into (i) two-stage frameworks and (ii) end-to-end frameworks. Two-stage frameworks generally first learn compressed representations via autoencoders, and then conduct DRP based on the compressed representations (Chiu *et al.*, 2019; Ding *et al.*, 2018). End-to-end frameworks learn high-level representations from raw data, and infer drug responses simultaneously. Specifically, MOLE (Sharifi-Noghabi *et al.*, 2019) used three type-specific sub-networks to learn representations from multi-omics data, and the learned representations were concatenated as input to the subsequent sub-network for DRP. CDRscan (Chang *et al.*, 2018) used Convolutional Neural Networks (CNNs) to learn representations of drugs and cell lines from molecular fingerprints and somatic mutation, respectively. Besides, CNNs have been utilized to process the SMILES (Liu *et al.*, 2019) and Kekule structures (Cort s-Ciriano and Bender, 2019) of drugs.

Exploiting the 2D topology of molecules, drugs can be explicitly represented as molecular graphs, whose nodes and edges denote atoms and chemical bonds, respectively. As deep learning-based methods have been developed for graph problems, GNNs have been applied to molecule representation learning and become state-of-the-art approaches for drug discovery (Sun *et al.*, 2020). More recently, attempts of DRP also paid close attention to processing drugs with GNNs. DeepCDR (Liu *et al.*, 2020) proposed a uniform graph convolutional network (UGCN) to capture the intrinsic chemical structures of drugs. However, few studies have been conducted on applying GNNs to learn the representations of cell lines.

We summarize the major limitations which existing deep learning-based DRP methods still suffer:

- Existing methods are not yet able to capture the relationships among genes which are essential for accurately representing cell lines. For example, CNNs have become a leading architecture in computer vision due to their ability of exploring spatial topology of images (Lecun *et al.*, 1998); but, CNNs are still unsuited for dealing with genomic profiles without readily exploitable spatial information.
- Existing methods have not taken advantage of the similarity among cell lines/drugs well enough. Note that cell lines with similar genetic profiles and drugs with similar chemical structures

generally have similar drug responses, as discussed in previous studies (Wang *et al.*, 2017; Zhang *et al.*, 2015).

- Existing methods still face difficulties in generalization to unseen cell lines and drugs in blind tests, which means that these methods are far insufficient for precision medicine in clinical cancer applications.

To address the limitations above, we propose a novel DRP framework, called Twin Graph neural networks with Similarity Augmentation (TGSA) (see Fig. 1 for an overview). TGSA is composed of Twin Graph neural networks for Drug Response Prediction (TGDRP) and a Similarity Augmentation (SA) module, incorporating prior domain knowledge to fuse fine-grained (gene-level and atom-level) and coarse-grained (sample-level) information. TGDRP utilizes two GNN encoders to learn the representations of drugs and cell lines based on molecular graphs and STRING protein-protein association networks (Szkarczyk *et al.*, 2019), respectively, and then the learned representations are concatenated and fed to a fully connected network (FCN) to make the final prediction. In the SA module, we incorporate similarity among cell lines/drugs by viewing DRP as an edge regression problem on a heterogeneous graph in which each node denotes either a drug or a cell line. SA utilizes GNNs to smooth the representations of similar cell lines/drugs. In addition, we introduce an auxiliary pre-training strategy for TGDRP's drug encoder to alleviate the issues of insufficient data and poor generalization on out-of-distribution molecules. Our main contributions are as follows.

- Better use of prior domain knowledge.** We propose TGDRP to incorporate relationships among genes, which are paramount for capturing complex patterns of cell lines. We also propose SA to incorporate similarity among drugs and cell lines. To the best of our knowledge, we are the first to utilize GNNs in DRP for cell line representation learning.
- Information fusion at different levels of granularity.** TGDRP and SA are coupled into TGSA to fuse fine-grained and coarse-grained information.
- A strategy for the dilemma of limited data.** We adapt an auxiliary pre-training strategy to further improve the generalization, especially on out-of-distribution drugs.
- Promising performance.** Our comprehensive experiments show that TGSA outperforms all the baselines on the GDSC2 dataset under various experimental settings empirically, demonstrating the superior predictive ability of TGSA.

2 Materials and methods

2.1 Problem definition

In this work, we formulate DRP as a regression problem for the log-normalized half maximal inhibitory concentration ($\ln(\text{IC}_{50})$) values of corresponding drug-cell line pairs with a mapping function $f: \mathcal{D} \times \mathcal{C} \rightarrow Y$, where $\mathcal{D} = \{d_1, d_2, \dots, d_n\}$ and $\mathcal{C} = \{c_1, c_2, \dots, c_m\}$ are the sets of drugs and cell lines, respectively, and $Y \in \mathbb{R}^{m \times n}$ is the drug response matrix, in which each entry Y_{ij} is the $\ln(\text{IC}_{50})$ value of c_i and d_j .

2.2 Data preparation

In this section, we describe how to integrate three public datasets, GDSC2, CCLE and COSMIC (Tate *et al.*, 2019), into our experimental dataset. Moreover, we show how to represent both drugs and cell lines as graphs.

Data integration. GDSC2 provides $\ln(\text{IC}_{50})$ values across hundreds of cell lines and drugs, and CCLE provides elaborate genetic profiles for cell lines, among which we take gene expression (EXP), somatic mutation (MU) and copy number variation (CNV) into account. Besides, 706 cancer-related genes are selected according to

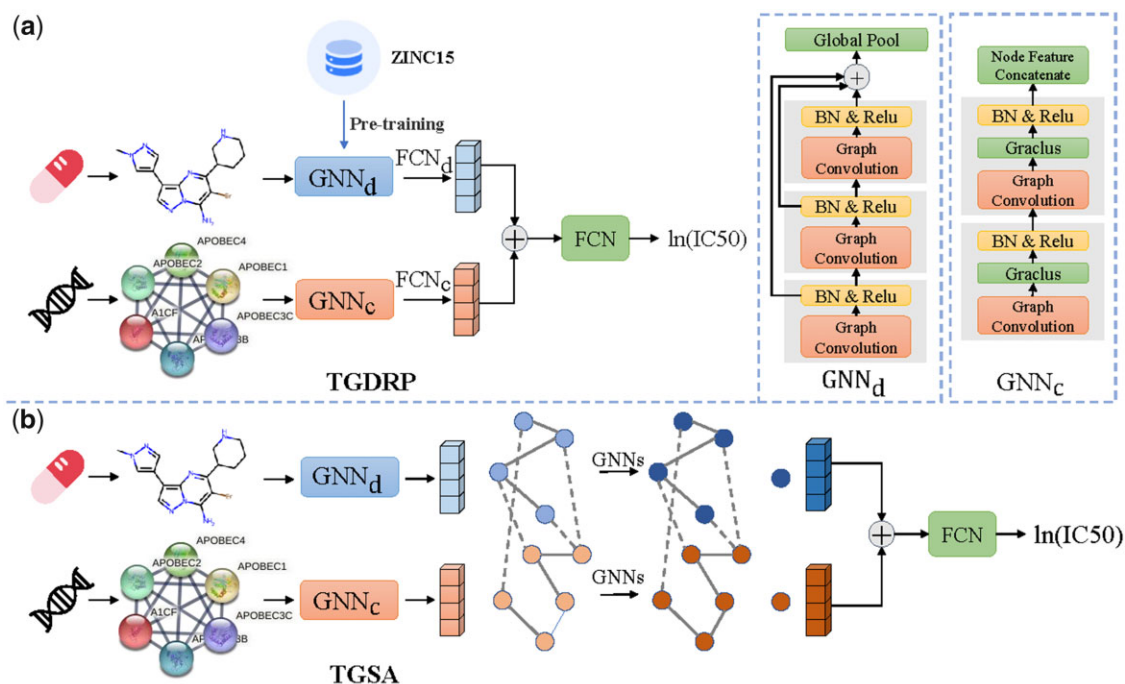


Fig. 1. An overview of our proposed TGSA framework. (a) TGD RP uses two GNNs to learn the drug and cell line representations, and then applies an FCN to predict $\ln(\text{IC}_{50})$. (b) TGSA uses TGD RP's GNN_d and GNN_c to generate the initialized features of the nodes in a heterogeneous graph. In the heterogeneous graph, the blue and orange nodes denote drugs and cell lines, respectively (the colors of the nodes changed from light to dark indicate an evolution of the node representations), and the solid and dashed edges denote similarity and drug-cell line responses, respectively

COSMIC (Tate *et al.*, 2019) as feature selection. Note that those drugs without PubChem ID and the cell lines missing any types of aforementioned omics data are removed. Consequently, we obtain 82833 available $\ln(\text{IC}_{50})$ across 580 cell lines and 170 drugs (see Supplementary Tables S1–S3), while approximately 16% (15767) of the $\ln(\text{IC}_{50})$ are unknown. A profile of the basic statistics of GDSC2 is shown in Supplementary Figure S1.

Graph construction. Generally, a graph $G = (\mathcal{V}, \mathcal{E})$ can be denoted as (A, F) , where $\mathcal{V}$ is a set of N nodes, $\mathcal{E}$ is a set of edges, $A \in \{0, 1\}^{N \times N}$ denotes the adjacency matrix, and $F \in \mathbb{R}^{N \times K}$ denotes matrix of node feature vectors. In our work, each molecular graph G^d is constructed following (Liu *et al.*, 2020), and the details of all the atom features are presented in Table 1. For each cell line graph $G^c = (\mathcal{V}^c, \mathcal{E}^c)$, $\mathcal{V}^c$ and $\mathcal{E}^c$ denote the sets of genes and interactions among them, respectively. $F^c \in \mathbb{R}^{706 \times 3}$ are determined by the aforementioned gene-wise multi-omics data. Unlike chemical bonds in molecules, there are no explicit relational structures among genes in cell lines. Thus, we introduce a gene interaction network as the prior knowledge to define $A^c \in \mathbb{R}^{706 \times 706}$. STRING is a curated dataset containing interactions among genes (Szklarczyk *et al.*, 2019). Formally, $A_{ij}^c = 1$ if the combined interaction score between gene i

and gene j in STRING is larger than a predefined threshold τ , set to 0.95 by default, and otherwise $A_{ij}^c = 0$. Statistically, G^c is a disconnected graph with the largest connected component of size 408, sparsity of 0.006 and average degree of 4.46.

2.3 Our proposed method

Without loss of generality, we first briefly review Message Passing Neural Networks (MPNNs) (Gilmer *et al.*, 2017) as a typical framework, to introduce the basic concepts of GNNs. Next, we present TGD RP and SA (Fig. 1). Finally, we discuss how to combine TGD RP and SA into TGSA.

MPNNs. The main idea of MPNNs is that each node can recursively receive messages from its neighbor nodes in the message passing stage. In the readout stage, MPNNs utilize a permutation-invariant readout function to summarize all node representations into fixed-length graph-level representations, which can be used for various downstream tasks. Formally, given a graph G , its node representations $b_v^l \in \mathbb{R}^m$ for node v and graph-level representations z^l in layer l can be computed as follows:

$$m_v^l = \sum_{w \in N(v)} M_{l-1}(b_v^{l-1}, b_w^{l-1}, e_{vw}), \quad (1)$$

$$b_v^l = U_{l-1}(b_v^{l-1}, m_v^l), \quad (2)$$

$$z^l = R(\{b_v^l \mid v \in G\}), \quad (3)$$

where m_v^l denotes the summed message of node v 's neighbors in layer $l-1$, $N(v)$ denotes the set of neighbors of node v , e_{vw} denotes the features of the edge between nodes v and w , and M , U and R are message function, update function and readout function, respectively.

TGD RP. In order to explore and exploit the topology of cell line graphs and better learn cell line representations, we propose the GNN-based TGD RP. As illustrated in Figure 1, TGD RP is a two-branch network that takes a molecular graph G^d and a cell line

Table 1. Atom features

Feature	Size	Description
Atom type	43	C, N, O, S, F, etc. (one-hot)
Degree	11	0–10 (one-hot)
Implicit valence	7	0–6 (one-hot)
Formal charge	1	Formal charge number (integer)
Radical electrons	1	Number of radical electrons (integer)
Hybridization	5	SP, SP2, SP3, SP3D, SP3D2 (one-hot or null)
Aromatic	1	Whether the atom is in an aromatic system (binary)
Hydrogens	5	0–4 (one-hot)
Ring	1	Whether the atom is in ring (binary)
Chirality	2	R, S (one-hot or null)

graph G^c as input and outputs the predicted $\ln(\text{IC}_{50})$. TGDRP comprises of the following three components.

Drug branch. Given a molecular graph G^d , we use the Graph Isomorphism Network (GIN) (Xu et al., 2019) as GNN_d to update atom features. GINs yielded state-of-the-art performance on a host of molecule-related tasks (Hu et al., 2020). Inspired by Jumping Knowledge Network to preserve information at different scales (Xu et al., 2018), we concatenate graph-level representations across layers which are read out by global max pooling, followed by FCN_d to learn the drug representations $z_d \in \mathbb{R}^{256}$.

Cell line branch. Given a cell line graph G^c , we use the Graph Attention Network (GAT) (Velickovic et al., 2018) as GNN_c to update gene features. During the message passing phase, GATs assign different weights to different neighbor nodes with a self-attention mechanism. Cell line graphs contain hierarchical information on genes' interactions (Erwin and Davidson, 2009). But, vanilla GAT and global pooling are inherently flat and unable to capture such information (Ying et al., 2018). Hence, we use Graclus (Dhillon et al., 2007) to coarsen the graph gradually by clustering two nodes into one 'super node' after every layer of GAT. Based on spectral clustering and kernel k-means, Graclus is an efficient and model-free graph clustering algorithm (Bianchi et al., 2020). A previous study demonstrated the effectiveness of Graclus in grouping the genes strongly connected (Dhillon et al., 2007). With this operation, not only the hierarchical structure of cell line graphs is captured, but also the advantages of pooling operations in CNNs are preserved. There are two beneficial characteristics of cell line graphs. (i) All the cell line graphs share the same topology. In practice, we only need to run Graclus once before training, to avoid duplicated computation. (ii) Cell line graphs are of larger scale compared to molecular graphs. We straightly concatenate all the super node representations to read out the graph-level representations, rather than global pooling which preserves only first-order statistics and loses much useful information. With the graph-level representations, FCN_c subsequently outputs the cell line representations $z_c \in \mathbb{R}^{256}$.

Prediction FCN. The drug and cell line representations are concatenated, and fed to the prediction FCN, to produce the final predicted $\ln(\text{IC}_{50})$.

Pre-training strategy. Pre-training strategies have been studied extensively in computer vision (He et al., 2019) and natural language processing (Kenton et al., 2019), but their effect in DRP has not been well explored. Training effective GNNs to capture the characteristics of drugs can be a great challenge due to the following two issues. (i) limited data: there are only 170 drugs in the whole dataset. (ii) Out-of-distribution samples: drug graphs in the test set are structurally quite different from those in the training set, leading to poor out-of-distribution generalization (Hu et al., 2020). To alleviate these issues, we pre-train GNN_d in large-scale molecule datasets and transfer the domain knowledge of chemistry to our task.

In this study, we follow the pre-training strategy in (Hu et al., 2020), whose key idea is to capture the domain-specific semantics at both the node and graph levels. Specifically, we first conduct node-level self-supervised pre-training on the ZINC15 dataset (Sterling and Irwin, 2015) with Deep Graph Infomax (DGI) (Velickovic et al., 2019). Then, we conduct graph-level multi-task supervised pre-training on the ChEMBL dataset (Gaulton et al., 2012), which contains 456k molecules with 1310 biochemical properties. During pre-training, GNN_d accumulates chemical knowledge of molecules (e.g. valency and biochemical properties). Unlike (Hu et al., 2020), we randomly hold out a validation set for model selection.

Similarity augmentation. As in the discussion of the second limitation of the known deep learning-based DRP methods in Section 1, similar cell lines/drugs tend to have similar representations, eliciting similar responses. Prior knowledge of cell line/drug similarity has been commonly considered as regularization terms to improve the DRP performance in matrix factorization methods (Guan et al., 2019; Wang et al., 2017). However, such regularization terms show poor extrapolation, and only attempt to reduce the difference of representations of similar cell lines/drugs, but ignore the extra relationship among similar cell lines/drugs. GNNs as low-pass filters can make the adjacent nodes in a graph tend to share similar

representations (Wu et al., 2019) and pass message between adjacent nodes (Gilmer et al., 2017). Therefore, we take advantage of the similarity among cell lines/drugs explicitly and sidestep above issues by resorting to GNNs. Specifically, we formulate DRP as an edge regression problem on a heterogeneous graph in which each node denotes either a drug or a cell line, and utilize GNNs to smooth the cell line/drug node representations. This is our Similarity Augmentation (SA) module.

The aforementioned heterogeneous graph is constructed by connecting two homogeneous k-nearest-neighbor graph of drugs and cell lines via known drug-cell line response. Formally, we define the heterogeneous graph as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $\mathcal{V} = \mathcal{D} \cup \mathcal{C}$ and $\mathcal{E} = \mathcal{E}_{\text{IC}} \cup \mathcal{E}_d \cup \mathcal{E}_c$. $\mathcal{E}_d$ and $\mathcal{E}_c$ are formed so that each drug node or cell line node is connected to its k (set to 5 by default) most similar neighbors according to the Jaccard similarity of Extended Connectivity Fingerprints (ECFP) (Rogers and Hahn, 2010) and Pearson correlation coefficient of gene expression, respectively. While $\mathcal{E}_{\text{IC}}$ denotes edges between drug nodes and cell line nodes with labels of corresponding $\ln(\text{IC}_{50})$.

For each cell line/drug, type-specific GraphSAGE (Hamilton et al., 2017) is applied to the heterogeneous graph $\mathcal{G}$ to update its representations with the message from its neighbors. In this way, informative similarity-aware node representations are obtained, and the obtained representations of adjacent nodes (similar cell lines/drugs) tend to be similar. The computing process of SA can be formulated as follows:

$$Z^c = \text{GraphSAGE}_c(X^c, \mathcal{E}_c) \quad (4)$$

$$Z^d = \text{GraphSAGE}_d(X^d, \mathcal{E}_d) \quad (5)$$

where $\text{GraphSAGE}_c(\cdot)$ and $\text{GraphSAGE}_d(\cdot)$ are type-specific GraphSAGE networks (the detailed formulas are shown in [Supplementary Material](#)) for cell lines and drugs, respectively. $X^c \in \mathbb{R}^{m \times \dim_c}$ and $X^d \in \mathbb{R}^{n \times \dim_d}$ are the initial feature matrices of cell lines and drugs, with each row corresponding to a cell line/drug. $Z^c \in \mathbb{R}^{m \times 256}$ and $Z^d \in \mathbb{R}^{n \times 256}$ are the output representation matrices of cell lines and drugs. After obtaining the cell line representations Z_i^c and the drug representations Z_j^d , the prediction step is the same as that of TGDRP.

TGSA. Because the nodes in TGDRP and in SA denote objects at different levels of granularity (i.e. genes versus cell lines and atoms versus drugs), TGDRP captures gene-level and atom-level information, while SA captures sample-level information. To fuse such fine-grained and coarse-grained information, we seek to couple TGDRP and SA via fine-tuning.

First, we train TGDRP end-to-end. Then in SA, X^d and X^c are generated by applying the corresponding trained GNN_d and GNN_c to the sets of molecular graphs and cell line graphs. To avoid additional parameters, the hidden dimensions and number of hidden layers of the type-specific GraphSAGEs and prediction FCN are kept the same as the corresponding FCNs in TGDRP. Finally, for fine-tuning, the parameters of GraphSAGEs and the prediction FCN are initialized by the corresponding trained FCNs in TGDRP. Through these processes, TGDRP can be viewed as a special case of TGSA with $\mathcal{E}_d \cup \mathcal{E}_c = \emptyset$.

2.4 Experimental setup

To comprehensively demonstrate the effectiveness of our method, we aim to examine the following three questions:

- Q1: Could TGSA outperform state-of-the-art baselines under diverse experimental settings?
- Q2: Does each component of TGSA work well?
- Q3: Could TGSA capture any biological meaning?

As suggested in (Baptista et al., 2021), we evaluate our method under two experimental settings. (i) For rediscovering known drug-cell line responses: The whole dataset is split into training/validation/test sets with a ratio of 8:1:1 by stratified sampling, and each

experiment is repeated with 10 random seeds. (ii) For blind test (leave-drug/cell-line-out): The whole dataset is split on the cell line/drug level to guarantee that the test set only includes unseen cell lines/drugs in the training stage, and 5-fold cross validation is conducted, where three folds are regarded as training set, and the other two folds are regarded as validation set and test set. This more rigorous scenario is in line with the clinical applications of drug repositioning and recommendation. Given that the data distribution of the test set may be quite different from that of the training set, cross validation can estimate the true generalization errors more accurately.

Three representative state-of-the-art methods: CDRscan (Chang *et al.*, 2018), MOL (Sharifi-Noghabi *et al.*, 2019) and tCNNS (Liu *et al.*, 2019) and two recent GNN-based methods: GraphDRP (Nguyen *et al.*, 2021) and DeepCDR (Liu *et al.*, 2020) are considered as our competing baselines. To explore the effectiveness of the cell line graph and GNN_c, we make some minor adjustments to DeepCDR for fair comparison. Specifically, UGCN of DeepCDR is replaced by our drug branch, and the multi-omics data of DeepCDR is replaced by ours as well. We denote this variant as DeepCDR*. Three widely used metrics for regression tasks are adopted to measure the performance: Root Mean Square Error (RMSE), Mean Absolute Error (MAE) and Pearson correlation coefficient (*r*).

To train our model, we use the mean square error as the loss function. All the baselines and our model are implemented in PyTorch (Paszke *et al.*, 2019) and PyTorch Geometric (Fey and Lenssen, 2019). The implementation details are shown in Supplementary Material. The hyper-parameters are tuned by grid search on the validation set (see Supplementary Table S4). We report the mean and standard deviation values, and conduct Student's *t*-test to analyze whether the performance difference is statistically significant between our method and the baselines.

3 Experimental results

3.1 Comparison with baselines

Table 2 compares the architectures and performance of our method with the state-of-the-art baselines. Our method significantly outperforms all the baselines on all the three metrics, providing positive answers to Q1. We discuss the results for diverse experiment settings as follows.

Rediscovering known drug-cell line responses. tCNNS and GraphDRP attain comparable performances, while GraphDRP is less robust with higher standard deviation values. This phenomenon is inconsistent with the observation that GNNs tend to outperform traditional methods on many molecule-related tasks (Sun *et al.*, 2020). We argue that it is difficult for GNNs to learn expressive and robust representations with such limited drugs. DeepCDR* surpasses other baselines significantly. We think that DRP benefits from multi-omics data, which is a main difference between DeepCDR* and GraphDRP. Thus, we suggest using as much related omics data as possible to represent cell lines, since such data can stably improve performance without increasing too much computation costs.

The performance improvement of TGDRP_{w/o pre} over DeepCDR* can be simply attributed to obtaining better cell line representations, which demonstrates that cell line graphs provide richer information than the raw multi-omics profiles and GNNs are capable of learning sophisticated patterns from cell line graphs.

Compared with DeepCDR*, TGSA attains a decrease of 0.037 in RMSE and 0.028 in MAE, which is about 4.1% relative improvement. Figure 2 shows the prediction results of TGSA. Specifically, Figure 2c illustrates the relationship between the observed ln(IC50) and predicted ln(IC50), which is highly linearly correlated. Figure 2a and b visualizes the distributions of the test RMSE on the cell line/drug level, which are approximate normal distributions. Compared with cell lines, the generalization on several drugs is particularly poor (RMSE > 3). This phenomenon might be due to the out-of-distribution test drugs in the enormous chemical space.

Blind test. As presented in Figure 2e and f, even though all the methods do not perform well in the blind test scenarios compared to the above scenario, especially in the leave-drug-out scenario, our method still outperforms the best baseline DeepCDR*. Note that we do not compare the performances of all the variants, because pre-training and SA only explicitly affect drug and cell line representations, respectively (as discussed in Section 3.2).

In leave-drug-out, the *r* of TGDRP_{w/o pre} is higher than DeepCDR* by 0.033 (0.493–0.46), and TGDRP achieves the best *r* of 0.527. Decreased performance indicates that our method still faces difficulties in generalizing to structurally different drugs. The pre-training strategy alleviates this defect by utilizing large-scaled molecules; nevertheless, the improvement is still unsatisfactory. We argue that this is because anti-cancer drugs are a very special kind of molecule and general domain knowledge seems insufficient. In leave-cell-line-out, TGDRP and TGSA are slightly better than DeepCDR (0.874–0.872).

Our experiments demonstrate that TGSA shows potential in the leave-cell-line-out scenario compared to leave-drug-out, indicating that the TGSA predictions could be taken as a reference for drug recommendation.

3.2 Ablation experiments

To answer Q2, we conduct ablation experiments from three aspects: gene features, the topology of cell line graphs and framework components. Note that we also pre-train DeepCDR* (DeepCDR_{pre}) for a fair comparison.

Gene features. For three types of omics data, namely, gene expression (EXP), somatic mutation (MU) and copy number variation (CNV), are used as gene features, it is of great necessity to evaluate the contributions of different types of omics data individually. Specifically, for each single-omics data, we exclude other types of omics data from the cell line graphs ($F^c \in \mathbb{R}^{706 \times 1}$). As shown in the upper half of Table 3 (the used single-omics data are denoted with subscript), TGDRP_{CNV} achieves the best performance among three single-omics variants, and TGDRP can even outperform

Table 2. Comparison of the model architectures and performances in the scenario of rediscovering known drug-cell line responses

Method	Input feature format		Feature encoder		RMSE	MAE	<i>r</i>
	Cell line	Drug	Cell line	Drug			
MOLI	Sequence	–	FCN	–	1.130 ± 0.027	0.875 ± 0.023	0.931 ± 0.005
CDRscan	Sequence	Fingerprint	CNN	CNN	0.993 ± 0.013	0.737 ± 0.011	0.937 ± 0.002
tCNNS	Sequence	SMILES	CNN	CNN	0.951 ± 0.009	0.700 ± 0.008	0.942 ± 0.001
GraphDRP	Sequence	Molecular graph	CNN	GNN	0.953 ± 0.020	0.702 ± 0.017	0.942 ± 0.003
DeepCDR*	Sequence	Molecular graph	FCN	GNN	0.914 ± 0.019	0.674 ± 0.016	0.946 ± 0.002
TGDRP _{w/o pre}	Graph	Molecular graph	GNN	GNN	0.894 ± 0.013	0.657 ± 0.010	0.949 ± 0.002
TGDRP	Graph	Molecular graph	GNN	GNN	0.885 ± 0.009	0.650 ± 0.006	0.950 ± 0.001
TGSA	Graph	Molecular graph	GNN	GNN	0.877 ± 0.008	0.646 ± 0.006	0.951 ± 0.001

^aWe report the standard deviation values after ‘±’. The best performance is highlighted in bold.

^bThe comparison of the model architectures is based on the perspectives of input feature format and feature encoder.

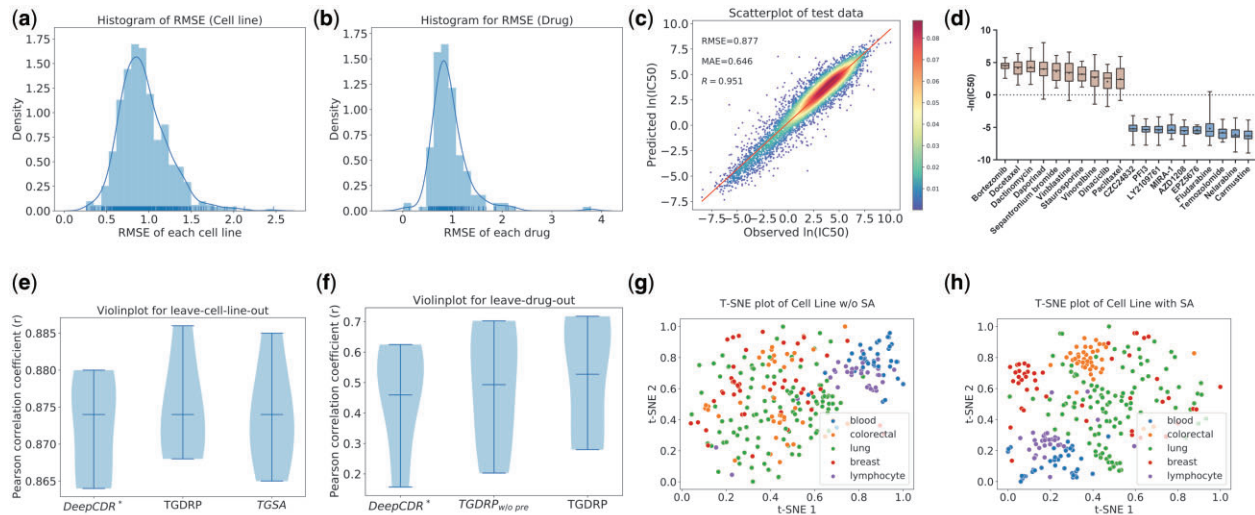


Fig. 2. Performance and comparison results. (a, b) The histograms illustrate the distribution of RMSE for each cell line/drug in the test set. (c) The scatter plot presents the observed $\ln(\text{IC}_{50})$ and predicted $\ln(\text{IC}_{50})$ in the test set. (d) The box plot presents the distribution of the predicted $\ln(\text{IC}_{50})$ of the missing drug-cell line pairs grouped by drugs. (e, f) The violin plots show performances in the leave-drug/cell-line-out scenarios. (g, h) The t-SNE plots present 2D representations of cell lines colored by corresponding tissue labels without/with SA

Table 3. Ablation studies

Method	RMSE	MAE	r
TGDRP _{EXP}	0.891 ± 0.009	0.655 ± 0.008	0.949 ± 0.001
TGDRP _{MU}	0.900 ± 0.018	0.661 ± 0.015	0.948 ± 0.002
TGDRP _{CNV}	0.889 ± 0.008	0.653 ± 0.007	0.950 ± 0.001
DeepCDR _{pre} *	0.904 ± 0.010	0.665 ± 0.008	0.948 ± 0.001
TGDRP _{Erdős-Rényi}	0.905 ± 0.008	0.665 ± 0.008	0.948 ± 0.001
TGDRP _{permutation}	0.900 ± 0.013	0.662 ± 0.012	0.948 ± 0.002
TGDRP _{pearson}	0.891 ± 0.015	0.656 ± 0.014	0.949 ± 0.002
TGDRP	0.885 ± 0.009	0.650 ± 0.006	0.950 ± 0.001

*We report the standard deviation values after ‘±’. The best performance is highlighted in bold.

DeepCDR_{pre}* with any single-omics data, indicating the effects of all single-omics data and the powerful capability of TGDRP.

Topology. To assess the effects of the STRING-based topology of cell line graphs, we evaluate the performance of TGDRP with diverse topological structures: random counterpart generated by Erdős-Rényi model (Erdos et al., 1960), permutational counterpart generated by permuting A^c , and statistic counterpart generated by replacing combined interaction score in STRING with Pearson correlation coefficient of gene expression in the process of graph construction. Note that the average degree of each counterpart is set to be the same as that of STRING-based topology for a fair comparison. As shown in the lower half of Table 3 (the used topology is denoted with subscript), TGDRP_{Erdős-Rényi} (TGDRP with meaning-less topology) achieves comparable performance with DeepCDR_{pre}*, and the STRING-based topology outperforms all counterparts, demonstrating that the improvement over DeepCDR_{pre}* comes from the beneficial prior knowledge contained in STRING.

Framework component. Pre-training strategies and SA have rarely been considered in previous DRP methods. To verify their effectiveness in our method, we conduct ablation experiments to compare two variants of TGSA: TGDRP_{w/o pre} (i.e. TGDRP without using the pre-training strategy) and TGDRP. Their performances are shown in the lower half of Table 2.

Considering our SA module, we find that the most decrease in RMSE (0.008) is attained only with cell line similarity augmentation, which indicates that drug similarity augmentation seems useless. This phenomenon is aligning with some recent research (Wang et al., 2017), while contrasted with Zhang et al. (2015). We doubt

that the drug response similarity is too complicated to be measured by Jaccard similarity of ECFP. Figure 2g and h visualize the t-SNE (Van der Maaten and Hinton, 2008) 2D representations of the cell lines from the top 5 most frequent tissue categories (without and with SA), which confirm the effect of SA since SA makes the cell lines from different tissues easier to distinguish. Likewise, visualization results colored by TCGA cancer type are shown in Supplementary Figure S2.

On our pre-training strategy, TGDRP’s RMSE decreases by 0.009. We point out an unexpected finding that considering only node-level pre-training can yield better performance than combining with graph-level pre-training. We speculate that such negative transfer may be caused by inconsistency between drug responses and the numerous and fuzzy biochemical assays in ChEMBL.

3.3 Biological meanings

A successful DRP method ought to not only make accurate predictions, but also capture important biological meanings. Hence, we further utilize a *post hoc* exploratory strategy and literature-based case studies to exhibit the capability of our method in discovering unknown drug-cell line responses and finding drug-gene interactions.

Discovering unknown drug-cell line responses. Among all the predicted $\ln(\text{IC}_{50})$ of the missing drug-cell line pairs (see Supplementary Table S5), the (Daporinad, EU-3) pair has the lowest value (-8.05) where EU-3 is a B-cell precursor leukemia cell line, indicating that Daporinad has excellent biological activities against hematologic malignant cells. This prediction is supported by Nahimana et al. (2009), which reported that most hematologic cancer cells are sensitive to low concentrations of Daporinad, on account of NAD depletion resulted from that Daporinad treatment elicits tumor cell death. Figure 2d illustrates the distributions of the predicted $\ln(\text{IC}_{50})$ values grouped by drugs which are ordered by the median predicted $\ln(\text{IC}_{50})$, and Bortezomib has the lowest median. This observation is aligned with the experimental results that Bortezomib has therapeutic effects on multiple cancers (Tan et al., 2019) such as relapsed multiple myeloma (Richardson et al., 2003) and breast cancer (Periyasamy-Thandavan et al., 2010).

Drug-gene interaction. Given a drug-cell line pair, we modify Grad-CAM (Selvaraju et al., 2017) to estimate the node-level gene importance since genes are represented as nodes in our method. Specifically, ReLU function is replaced with absolute function to be suited for our regression problem which is not only interest in the features that have a positive influence on $\ln(\text{IC}_{50})$, as the vanilla

Grad-CAM was proposed for classification problem. Accordingly, the formula of Grad-CAM can be rewritten as:

$$L[l, n] = \left| \sum_k b_{k,n}^l \sum_{n=1}^N \frac{\partial y}{\partial b_{k,n}^l} \right| \quad (6)$$

where $|\cdot|$ calculates the absolute value element-wise, y is the predicted $\ln(\text{IC}_{50})$, $b_{k,n}^l$ is the k 'th feature of the graph convolutional activations for node n in layer l , and $L[l, n]$ represents the importance score of node n in the corresponding graph.

According to the computed gene importance scores, the top 5 important genes of several cases are shown in Table 4. Remarkably, we notice that many top-ranked genes thus determined have been verified to be associated with the mechanism of action of the corresponding drugs. For instance, PAFAH1B2 is a target gene of HIF-1 α (Ma *et al.*, 2018), the latter of which has been reported to be inhibited by Bortezomib thus inhibiting tumor adaptation to hypoxia (Shin *et al.*, 2008). PAFAH1B2 ranks first with respect to the (Bortezomib, 8MGBA) pair. Moreover, Irinotecan is a derivative of Camptothecin targeting DNA replication pathway (Pommier, 2006). And in their sensitivity to KMH2 cell line, the target gene TOP1 and a DNA replication-related gene POLQ (DNA polymerase θ) rank fourth and third, respectively. Also, BRD4 has been reported to be a potential target of AZD5153 by regulating transcriptional programs (Rhyasen *et al.*, 2016) and was estimated by our method to be important in AZD5153 sensitivity to NCIH526 cell line.

Conclusively, these literature-based case studies exemplify that our TGSA could be applied to discover new therapeutic effects as well as potential therapeutic targets of anti-cancer drugs.

4 Discussion

In this article, we proposed a novel DRP framework, called TGSA, which incorporates prior domain knowledge to fuse fine-grained and coarse-grained information. To explore relationships among genes, we incorporated STRING protein-protein association networks to construct cell line graphs. To capture similarity among cell lines/drugs, we treated DRP as an edge regression problem on a heterogeneous graph, and utilized GNNs to smooth the representations of similar cell lines/drugs. Besides, we introduced an auxiliary pre-training strategy to alleviate the dilemma of limited data and improve the out-of-distribution generalization for drugs. Extensive experiments on the GDSC2 dataset demonstrated that our TGSA outperforms several state-of-the-art DRP methods under different experimental settings. Ablation study verified the effectiveness and contributions of each component of TGSA. Moreover, biological case analysis demonstrated that TGSA can, to some extent, enlighten experts to further explore related domain knowledge.

Despite the above promising performances, our method still needs to address several limitations and also reveals insight for future research. First, the topology of cell line graphs is constructed simply based on the combined interaction score of STRING. To determine graph topology dedicatedly, taking more abundant domain knowledge into account is a fruitful avenue. Second, Graclus only considers the graph topology while ignoring the node features (Bianchi *et al.*, 2020), and thus cell line graphs cannot be adaptively coarsened based on a specific DRP task. It is worth considering to apply feature-based hierarchical pooling algorithms to help GNNs

learn more sophisticated cell line representations. Finally, the performances of the blind test scenarios are still not satisfactory, especially the leave-drug-out scenario. Further efforts are needed to improve the out-of-distribution generalization of GNNs, and investigating pharmacophore-aware self-supervised learning appears promising for it.

To sum up, our TGSA integrates chemical domain knowledge into deep learning, and the promising performance of TGSA shows enormous potential for clinical applications in precision medicine.

Acknowledgements

The authors thank Minshan Lai and Yufeng Xie for their helpful discussions.

Funding

This work was supported by the National Research and Development Program of China [2019YFC0118802]; and the Key R & D Program of Zhejiang Province [2020C03010]; and the National Natural Science Foundation of China [61672453]; and the Zhejiang University Education Foundation under grants [K18-511120-004, K17-511120-017 and K17-518051-02]; and the Zhejiang public welfare technology research project [LGF20F020013]; and the Medical and Health Research Project of Zhejiang Province of China [2019KY667]; and the Wenzhou Bureau of Science and Technology of China [Y2020082]; and the Key Laboratory of Medical Neurobiology of Zhejiang Province. D.Z.C.'s research was supported in part by the National Science Foundation [CCF-1617735].

Conflict of Interest: none declared.

References

- Ammad-Ud Din, M. *et al.* (2016) Drug response prediction by inferring pathway-response associations with kernelized Bayesian matrix factorization. *Bioinformatics*, **32**, i455–i463.
- Baptista, D. *et al.* (2021) Deep learning for drug response prediction in cancer. *Brief. Bioinf.*, **22**, 360–379.
- Barretina, J. *et al.* (2012) The cancer cell line encyclopedia enables predictive modelling of anticancer drug sensitivity. *Nature*, **483**, 603–607.
- Bianchi, F.M. *et al.* (2020) Spectral clustering with graph neural networks for graph pooling. In: *ICML*. PMLR, Vienna, Austria, pp. 874–883.
- Chang, Y. *et al.* (2018) Cancer drug response profile scan (CDRscan): a deep learning model that predicts drug effectiveness from cancer genomic signature. *Sci. Rep.*, **8**, 8857–8811.
- Chiu, Y.-C. *et al.* (2019) Predicting drug response of tumors from integrated genomic profiles by deep neural networks. *BMC Med. Genomics*, **12**, 119–155.
- Cortés-Ciriano, I. and Bender, A. (2019) KekuleScope: prediction of cancer cell line sensitivity and compound potency using convolutional neural networks trained on compound images. *J. Cheminf.*, **11**, 41–16.
- Dhillon, J.S. *et al.* (2007) Weighted graph cuts without eigenvectors: a multi-level approach. *IEEE Trans. Pattern Anal. Mach. Intell.*, **29**, 1944–1957.
- Ding, M.Q. *et al.* (2018) Precision oncology beyond targeted therapy: combining omics data with machine learning matches the majority of cancer cells to effective therapeutics. *Mol. Cancer Res.*, **16**, 269–278.
- Dong, Z. *et al.* (2015) Anticancer drug sensitivity prediction in cell lines from baseline gene expression through recursive feature selection. *BMC Cancer*, **15**, 489–412.
- Erdos, P. *et al.* (1960) On the evolution of random graphs. *Publ. Math. Inst. Hung. Acad. Sci.*, **5**, 17–60.
- Erwin, D.H. and Davidson, E.H. (2009) The evolution of hierarchical gene regulatory networks. *Nat. Rev. Genet.*, **10**, 141–148.
- Fey, M. and Lenssen, J.E. (2019) Fast graph representation learning with PyTorch geometric. In: *ICLR Workshop, New Orleans, Louisiana, United States*.
- Friedman, A.A. *et al.* (2015) Precision medicine for cancer with next-generation functional diagnostics. *Nat. Rev. Cancer*, **15**, 747–756.
- Gaulton, A. *et al.* (2012) ChEMBL: a large-scale bioactivity database for drug discovery. *Nucleic Acids Res.*, **40**, D1100–D1107.
- Gilmer, J. *et al.* (2017) Neural message passing for quantum chemistry. In: *ICML*. PMLR, Sydney, Australia, pp. 1263–1272.

Table 4. Top 5 important genes with respect to drug-cell line pairs

Drug	Cell line	Top 5 important genes
Bortezomib	8MGBA	PAFAH1B2, DDX10, FAT3, FOXR1, KCNJ5
Camptothecin	KMH2	NUTM2B, FKBP9, POLQ, TOP1, CCX6C
Irinotecan	KMH2	FKBP9, NUTM2B, POLQ, TOP1, CCX6C
AZD5153	NCIH526	ELK4, HERPUD1, KLF6, RGS7, BRD4

Note: Gene importance is estimated based on the modified Grad-CAM method.

- Guan,N.-N. *et al.* (2019) Anticancer drug response prediction in cell lines using weighted graph regularized matrix factorization. *Mol. Therapy Nucleic Acids*, **17**, 164–174.
- Hamilton,W.L. *et al.* (2017) Inductive representation learning on large graphs. In: *NeurIPS*, Long Beach, CA, United States, pp. 1024–1034.
- He,K. *et al.* (2019) Rethinking ImageNet pre-training. In: *ICCV, Seoul, Korea*.
- Hodson,R. (2016) Precision medicine. *Nature*, **537**, S49.
- Hu,W. *et al.* (2020) Strategies for pre-training graph neural networks. In: *ICLR, Addis Ababa, Ethiopia*.
- Iorio,F. *et al.* (2016) A landscape of pharmacogenomic interactions in cancer. *Cell*, **166**, 740–754.
- Kenton,J. *et al.* (2019) BERT: pre-training of deep bidirectional transformers for language understanding. In: *NAACL-HLT*, Minneapolis, Minnesota, 4171–4186.
- Lecun,Y. *et al.* (1998) Gradient-based learning applied to document recognition. *Proc. IEEE*, **86**, 2278–2324.
- Liu,P. *et al.* (2019) Improving prediction of phenotypic drug response on cancer cell lines using deep convolutional network. *BMC Bioinformatics*, **20**, 408–414.
- Liu,Q. *et al.* (2020) DeepCDR: a hybrid graph convolutional network for predicting cancer drug response. *Bioinformatics*, **36**, i911–i918.
- Ma,C. *et al.* (2018) PAFAH1B2 is a HIF1a target gene and promotes metastasis in pancreatic cancer. *Biochem. Biophys. Res. Commun.*, **501**, 654–660.
- Mayr,A. *et al.* (2016) DeepTox: toxicity prediction using deep learning. *Front. Environ. Sci.*, **3**, 80.
- Nahimana,A. *et al.* (2009) The NAD biosynthesis inhibitor APO866 has potent antitumor activity against hematologic malignancies. *Blood*, **113**, 3276–3286.
- Nguyen,T.-T. *et al.* (2021) Graph convolutional networks for drug response prediction. *IEEE/ACM Trans. Comput. Biol. Bioinf.*
- Paszke,A. *et al.* (2019) PyTorch: an imperative style, high-performance deep learning library. In: *NeurIPS*, Vancouver, Canada, pp. 8024–8035.
- Periyasamy-Thandavan,S. *et al.* (2010) Bortezomib blocks the catabolic process of autophagy via a cathepsin-dependent mechanism, affects endoplasmic reticulum stress, and induces caspase-dependent cell death in antiestrogen-sensitive and resistant ER+ breast cancer cells. *Autophagy*, **6**, 19–35.
- Pommier,Y. (2006) Topoisomerase I inhibitors: camptothecins and beyond. *Nat. Rev. Cancer*, **6**, 789–802.
- Rhyasen,G.W. *et al.* (2016) Azd5153: a novel bivalent bet bromodomain inhibitor highly active against hematologic malignancies. *Mol. Cancer Therap.*, **15**, 2563–2574.
- Richardson,P.G. *et al.* (2003) A phase 2 study of bortezomib in relapsed, refractory myeloma. *N. Engl. J. Med.*, **348**, 2609–2617.
- Rogers,D. and Hahn,M. (2010) Extended-connectivity fingerprints. *J. Chem. Inf. Model.*, **50**, 742–754.
- Selvaraju,R.R. *et al.* (2017) Grad-cam: visual explanations from deep networks via gradient-based localization. In: *Proceedings of the IEEE International Conference on Computer Vision*, Venice, Italy, pp. 618–626.
- Sharifi-Noghabi,H. *et al.* (2019) MOLI: multi-omics late integration with deep neural networks for drug response prediction. *Bioinformatics*, **35**, i501–i509.
- Shin,D.H. *et al.* (2008) Bortezomib inhibits tumor adaptation to hypoxia by stimulating the FIH-mediated repression of hypoxia-inducible factor-1. *Blood. J. Am. Soc. Hematol.*, **111**, 3131–3136.
- Sterling,T. and Irwin,J.J. (2015) ZINC 15–ligand discovery for everyone. *J. Chem. Inf. Model.*, **55**, 2324–2337.
- Sun,M. *et al.* (2020) Graph convolutional networks for computational drug development and discovery. *Brief. Bioinf.*, **21**, 919–935.
- Suphavitai,C. *et al.* (2018) Predicting cancer drug response using a recommender system. *Bioinformatics*, **34**, 3907–3914.
- Szklarczyk,D. *et al.* (2019) STRING v11: protein–protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets. *Nucleic Acids Res.*, **47**, D607–D613.
- Tan,C.R.C. *et al.* (2019) Clinical pharmacokinetics and pharmacodynamics of bortezomib. *Clin. Pharmacokinet.*, **58**, 157–168.
- Tate,J.G. *et al.* (2019) COSMIC: the catalogue of somatic mutations in cancer. *Nucleic Acids Res.*, **47**, D941–D947.
- Teschendorff,A.E. (2019) Avoiding common pitfalls in machine learning omic data science. *Nat. Mater.*, **18**, 422–427.
- Van der Maaten,L. and Hinton,G. (2008) Visualizing data using t-SNE. *J. Mach. Learn. Res.*, **9**, 2579–2605.
- Velickovic,P. *et al.* (2018) Graph attention networks. In: *ICLR*, Vancouver, Canada.
- Velickovic,P. *et al.* (2019) Deep graph infomax. In: *ICLR*, New Orleans, Louisiana, United States.
- Wang,L. *et al.* (2017) Improved anticancer drug response prediction in cell lines using matrix factorization with similarity regularization. *BMC Cancer*, **17**, 513–512.
- Wu,F. *et al.* (2019) Simplifying graph convolutional networks. In *ICML*. PMLR, Long Beach, California, pages 6861–6871.
- Xu,K. *et al.* (2018) Representation learning on graphs with jumping knowledge networks. In: *ICML*. PMLR, Stockholm, Sweden, pp. 5453–5462.
- Xu,K. *et al.* (2019) How powerful are graph neural networks? In: *ICLR, New Orleans, Louisiana, United States*.
- Yang,W. *et al.* (2013) Genomics of drug sensitivity in cancer (GDSC): a resource for therapeutic biomarker discovery in cancer cells. *Nucleic Acids Res.*, **41**, D955–D961.
- Ying,Z. *et al.* (2018) Hierarchical graph representation learning with differentiable pooling. In: *NeurIPS*, Montréal, Canada.
- Zhang,N. *et al.* (2015) Predicting anticancer drug responses using a dual-layer integrated cell line-drug network model. *PLoS Comput. Biol.*, **11**, e1004498.